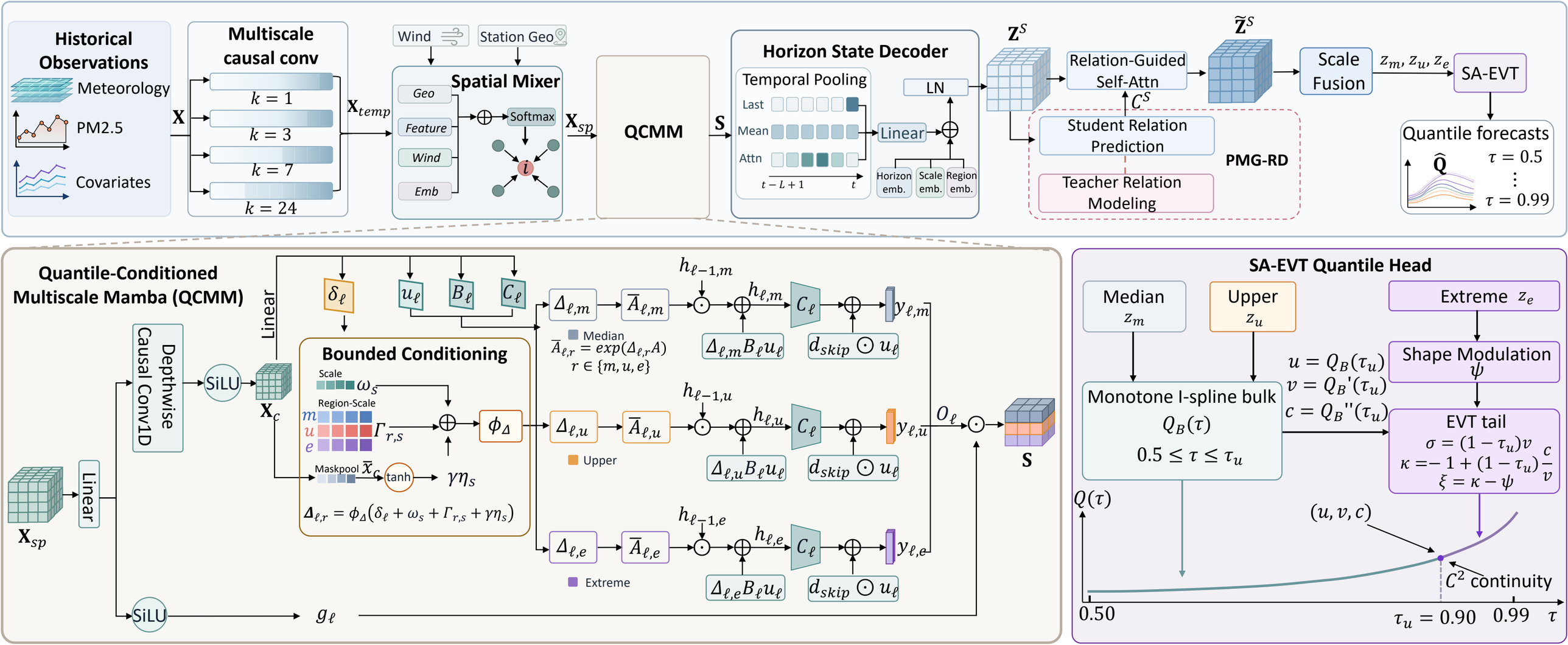

Historical Observations
Meteorology
PM2.5
Covariates
X
Multiscale causal conv
k = 1
k = 3
k = 7
k = 24
X_temp
Wind
Station Geo
Spatial Mixer
Geo
Feature
Wind
Emb
Softmax
i
X_sp
QCMM
S
Horizon State Decoder
Temporal Pooling
Last
Mean
Attn
t − L + 1
t
Linear
LN
Horizon emb.
Scale emb.
Region emb.
Z^S
Relation-Guided Self-Attn
C^S
Student Relation Prediction
Teacher Relation Modeling
PMG-RD
Z̃^S
Scale Fusion
z_m, z_u, z_e
SA-EVT
Quantile forecasts
Q̂
τ = 0.5
τ = 0.99
Quantile-Conditioned Multiscale Mamba (QCMM)
Linear
Depthwise Causal Conv1D
SiLU
X_c
X_sp
Linear
SiLU
g_ℓ
δ_ℓ
u_ℓ
B_ℓ
C_ℓ
Bounded Conditioning
Scale
ω_s
Region-Scale
m
u
e
Γ_{r,s}
Maskpool x̄_c
tanh
γη_s
ϕ_Δ
Δ_{ℓ,r} = ϕ_Δ(δ_ℓ + ω_s + Γ_{r,s} + γη_s)
Δ_{ℓ,m}
Ā_{ℓ,m}
Median
Ā_{ℓ,r} = exp(Δ_{ℓ,r}A)
r ∈ {m, u, e}
h_{ℓ−1,m}
h_{ℓ,m}
Δ_{ℓ,m}B_ℓu_ℓ
d_skip ⊙ u_ℓ
y_{ℓ,m}
Δ_{ℓ,u}
Ā_{ℓ,u}
Upper
h_{ℓ−1,u}
h_{ℓ,u}
Δ_{ℓ,u}B_ℓu_ℓ
y_{ℓ,u}
Δ_{ℓ,e}
Ā_{ℓ,e}
Extreme
h_{ℓ−1,e}
h_{ℓ,e}
Δ_{ℓ,e}B_ℓu_ℓ
y_{ℓ,e}
O_ℓ
S
SA-EVT Quantile Head
Median
z_m
Upper
z_u
Extreme z_e
Shape Modulation
ψ
u = Q_B(τ_u)
v = Q_B'(τ_u)
c = Q_B''(τ_u)
Monotone I-spline bulk
Q_B(τ)
0.5 ≤ τ ≤ τ_u
EVT tail
σ = (1 − τ_u)v
κ = −1 + (1 − τ_u) c/v
ξ = κ − ψ
Q(τ)
(u, v, c)
C² continuity
0.50
τ_u = 0.90
0.99
τ